



Computer Vision and Image Processing

Lecture

Image Segmentation

- **Image Segmentation** is a fundamental process in **Digital Image Processing and Computer Vision** that involves **dividing an image into multiple meaningful regions or segments**. Each segment corresponds to an object or a part of an object in the image.
- The primary objective is to **simplify or change the representation of an image** so that it becomes **easier to analyse and interpret**.

Image Segmentation

Image segmentation is a **pre-processing step** for higher-level vision tasks such as:

- Object detection and recognition
- Medical image analysis (tumour detection)
- Remote sensing and satellite imagery
- Autonomous driving
- Video surveillance

Image Segmentation

All segmentation techniques are based on **two fundamental image properties**:

1. Discontinuity

- Detects **abrupt changes** in pixel intensity
- Leads to **edge-based segmentation**

2. Similarity

- Groups pixels with **similar properties**
- Leads to **region-based and colour-based segmentation**

Image Segmentation

Basic Image Segmentation Techniques:

1. Edge-Based Segmentation
2. Colour-Based Segmentation
3. Region-Based Segmentation

Edge-Based Segmentation

Edge-based segmentation partitions an image by **detecting boundaries**—locations where there is a **significant discontinuity** in intensity or colour. These discontinuities typically correspond to **object borders**, making edges a natural cue for segmentation when region interiors are relatively uniform.



(a) Original



(b) Canny,
 $\sigma = 1$



(c) Canny,
 $\sigma = 3$

Edge-Based Segmentation

Roberts Operator

Uses 2×2 masks.

Very sensitive to noise; fast but coarse.

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Prewitt Operator

Uses 3×3 masks; uniform weights.

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Edge-Based Segmentation

Sobel Operator

Similar to Prewitt but emphasizes central pixels (better noise robustness).

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Edge-Based Segmentation

Canny Edge Detector

A multi-stage, optimal detector:

1. Gaussian smoothing (noise reduction)
2. Gradient computation
3. Non-maximum suppression
4. Double thresholding
5. Edge tracking by hysteresis

Provides thin, well-connected edges with low false detections.

Edge-Based Segmentation

Advantages

- Precise localization of object boundaries
- Computationally efficient (especially gradient-based methods)
- Effective when object borders are well-defined

Limitations

- Highly sensitive to noise (unless smoothed)
- Produces **edges**, **not regions**—additional processing is required for full segmentation
- Broken or spurious edges in textured or low-contrast images

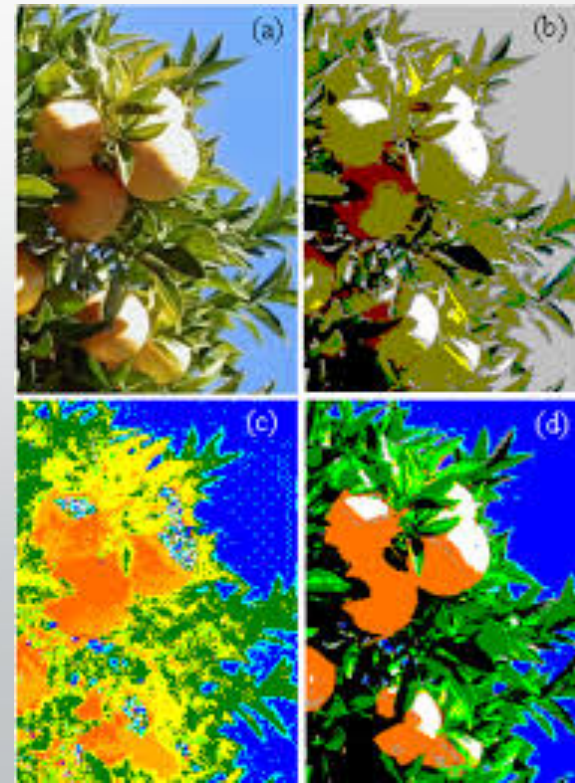
Edge-Based Segmentation

Applications

- Boundary detection in medical imaging
- Industrial inspection (crack detection)
- Document analysis
- Pre-processing for shape-based object recognition

Colour-Based Segmentation

Colour-based segmentation partitions an image by grouping pixels that are **similar in colour**. Unlike edge-based methods, which rely on intensity discontinuities, colour-based segmentation exploits the fact that **objects often have distinctive and consistent colour distributions**, making colour a powerful discriminative feature.



Colour-Based Segmentation

Basic Principle

Pixels are assigned to the same segment if their **colour distance** is within a threshold.

$$\|\mathbf{p} - \mathbf{c}\| \leq T$$

Where:

- $\mathbf{p}$ = pixel color vector
- $\mathbf{c}$ = reference color (cluster center)
- T = threshold

Colour-Based Segmentation

Distance Measures

Euclidean Distance (Most Common)

$$D = \sqrt{(R - R_c)^2 + (G - G_c)^2 + (B - B_c)^2}$$

Manhattan Distance

$$D = |R - R_c| + |G - G_c| + |B - B_c|$$

Colour-Based Segmentation

Input RGB Image (2×2)

$$I = \begin{bmatrix} (255, 0, 0) & (250, 10, 10) \\ (0, 255, 0) & (10, 250, 10) \end{bmatrix}$$

Reference Color (Red)

$$\mathbf{c} = (255, 0, 0)$$

Distance Calculation

Pixel (250,10,10):

$$\begin{aligned} D &= \sqrt{(250 - 255)^2 + (10 - 0)^2 + (10 - 0)^2} \\ &= \sqrt{25 + 100 + 100} = 15 \end{aligned}$$

Pixel (0,255,0):

$$D = \sqrt{(0 - 255)^2 + (255 - 0)^2 + (0 - 0)^2} \approx 361$$

Assume $T = 30$.

Segmented Output (Binary Mask)

$$S = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Pixels labeled 1 belong to the red segment.

Colour-Based Segmentation

Advantages

- Effective for images with **distinct colours**
- Simple mathematical formulation
- Works well for **multi-object scenes**
- Robust when using HSV or Lab spaces

Limitations

- Sensitive to illumination in RGB
 - Poor performance when object and background share similar colours
- Shadows and highlights can cause misclassification

Colour-Based Segmentation

Applications

- Object tracking and recognition
- Medical image analysis (tissue segmentation)
- Satellite and aerial image processing
- Traffic sign and lane detection

Region-Based Segmentation

Region-based segmentation partitions an image into regions by **grouping neighbouring pixels that are similar** according to a predefined homogeneity criterion (intensity, colour, texture).

Unlike edge-based segmentation, which focuses on **boundaries**, region-based segmentation focuses on **region interiors** and produces **closed, connected regions directly**.

Region-Based Segmentation

The fundamental idea is **similarity + spatial connectivity**.

Mathematically, a region R satisfies:

$$|I(x, y) - \mu_R| \leq T \quad \forall (x, y) \in R$$

Where:

- $I(x, y)$ = pixel intensity (or color vector)
- μ_R = mean value of the region
- T = similarity threshold

Region-Based Segmentation

Why Region-Based Segmentation Is Needed

- Edge-based methods:
- Detect boundaries only
- Often produce broken edges

Region-based methods:

- Produce **complete object regions**
- Are more robust in **low-contrast or noisy images**
- Are well-suited for **medical, satellite, and document images**

Region-Based Segmentation

Main Types of Region-Based Segmentation

- 1. Region Growing**
- 2. Region Splitting**
- 3. Region Merging**
- 4. Split-and-Merge (Hybrid)**

Region Growing

Region growing starts from **seed pixels** and **grows regions outward** by adding neighbouring pixels that are similar.

Algorithm Steps

1. Select one or more **seed pixels**
2. Initialize region with seed
3. Compare neighbouring pixels with region mean
4. Add pixel if similarity condition is satisfied
5. Update region mean

Repeat until no more pixels qualify

Region Growing

Input Image

$$I = \begin{bmatrix} 50 & 52 & 80 \\ 49 & 51 & 82 \\ 48 & 50 & 85 \end{bmatrix}$$

Seed Pixel

$$(0, 0) = 50$$

Threshold:

$$T = 3$$

Step-by-Step Growth

- Neighbor (0,1) = 52 $\rightarrow |52 - 50| = 2 \leq 3$ ✓
- Neighbor (1,0) = 49 $\rightarrow |49 - 50| = 1 \leq 3$ ✓
- Neighbor (1,1) = 51 $\rightarrow |51 - 50| = 1 \leq 3$ ✓
- Neighbor (0,2) = 80 $\rightarrow |80 - 50| = 30$ ✗
- Neighbor (1,2) = 82 $\rightarrow$ ✗

Final Region Mask

$$R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Region Growing

Advantages

- Produces connected regions
- Intuitive and simple

Limitations

- Sensitive to seed selection
- Computationally expensive for large images

Region Splitting

Concept

- Region splitting follows a **top-down approach**:
- Start with the whole image
- Split regions if homogeneity condition fails
- Continue recursively (usually into quadrants)

Region Splitting

Input Image

$$I = \begin{bmatrix} 10 & 10 & 80 & 80 \\ 10 & 10 & 80 & 80 \\ 12 & 12 & 82 & 82 \\ 12 & 12 & 82 & 82 \end{bmatrix}$$

Analysis

- Entire image variance → high ✗
- Split into 4 quadrants:

$$Q_1 = \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix} \quad Q_2 = \begin{bmatrix} 80 & 80 \\ 80 & 80 \end{bmatrix}$$

Each quadrant variance → low ✓

Region Splitting

Why Splitting Works

- Large regions often contain multiple objects
- Splitting isolates homogeneous subregions automatically

Limitations

- Produces block-shaped regions
- Can lead to over-segmentation

Region Merging

Region merging is a **bottom-up approach**:

- Start with many small regions
- Merge adjacent regions if they are similar

$$|\mu_{R_1} - \mu_{R_2}| \leq T$$

Region Merging

Two Adjacent Regions

$$R_1 = \begin{bmatrix} 50 & 51 \\ 49 & 50 \end{bmatrix} \Rightarrow \mu_{R_1} = 50$$

$$R_2 = \begin{bmatrix} 52 & 51 \\ 50 & 53 \end{bmatrix} \Rightarrow \mu_{R_2} = 51.5$$

Threshold:

$$T = 3$$

$$|50 - 51.5| = 1.5 \leq 3 \Rightarrow \text{Merge}$$

Region Merging

Why Merging Is Needed

- Reduces over-segmentation
- Combines meaningful regions
- Smooths noisy boundaries

Split-and-Merge (Hybrid Approach)

Concept

- First **split** regions until homogeneous
- Then **merge** adjacent similar regions

This balances:

- Over-segmentation (splitting)
- Under-segmentation (merging)



The End